SWE304 PROJECT STUDY 3 (2026)

Due date: 30 April 2026, class time.

In this project, you are expected to create a CI/CD pipeline on Github using workflow utility see Figure 1). Follow the Github action specifications for the project given below:

- A cloud instance must be created with ports 22, 80 open.
- Github action triggering events are a "**developer push**" or a "**pull request resulted with merge**".
- A new jar of web application must be created automatically on Github when an event-trigger occurs.
- Software versions (such as Java and build tool) must comply with your application.
- A script file on GitHub (not in the local) must be transferred (or copied) to the cloud instance.
- Docker image must be created on Github
- Dockerhub login must be done automatically using Github secrets.
- The docker image must be pushed to the Dockerhub.
- The image must be pulled from Dockerhub running the script on ec2 triggered by GitHub.

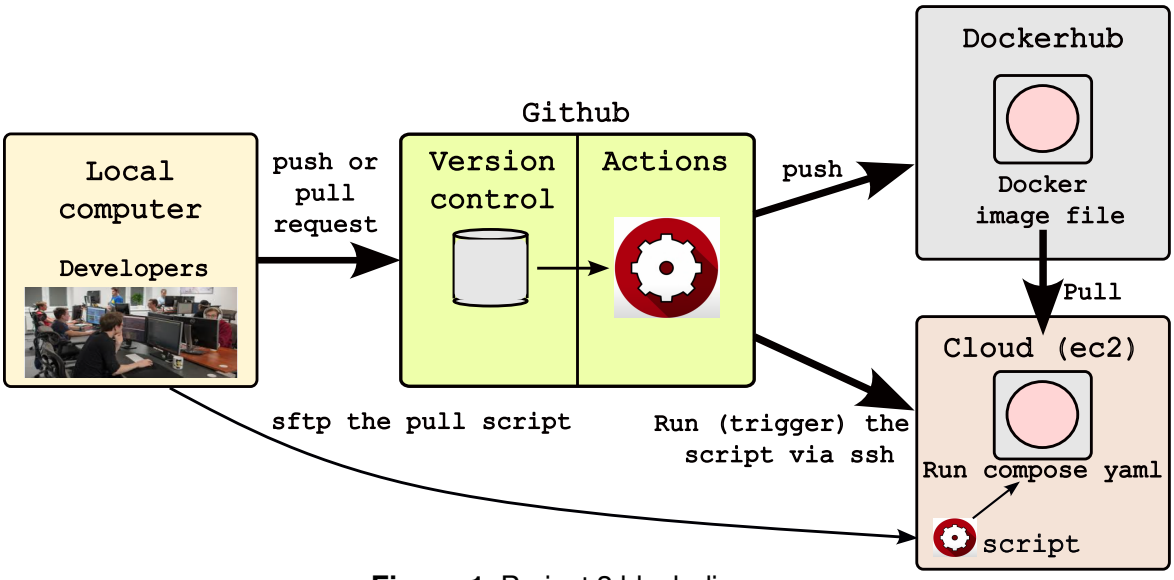

Figure 1. Project 3 block diagram.

PS: You must show that new version of your application is working automatically on the cloud after "push" or "merge" action on GitHub.

Grading:

No	Tasks (Jenkins labeled tasks must be done by Jenkins server)	Grade
1	Have your local webapp ready for update & push or pull request to GitHub repository.	10
2	Create workflow action script in your workflow for making JAR on GitHub.	20
3	Add an action script to your workflow to generate a docker image on GitHub.	20
4	Add action script to your workflow to login to Dockerhub and push the image.	20
5	Add action script to your workflow to trigger/run the compose.yml file.	20
6	Show that workflow file and CI/CD pipeline works as expected.	10

PS: Student must show up for presentation to collect points.